Soil Health & Crop Recommendation Report

Report Date: November 20, 2025 at 22:54:24 UTC

Farmer Information	
Name:	nandish
Phone:	8088117817
Location:	vijayapura
Land Size:	1.00 acres

Parameter	Value	Unit
Nitrogen (N)	50.0	mg/kg
Phosphorus (P)	50.0	mg/kg
Potassium (K)	50.0	mg/kg
pH Level	6.50	
Temperature	25.0	°C
Humidity	60.0	%
Rainfall	120.0	mm
Soil Type	Black Soil	

Soil Health Assessment

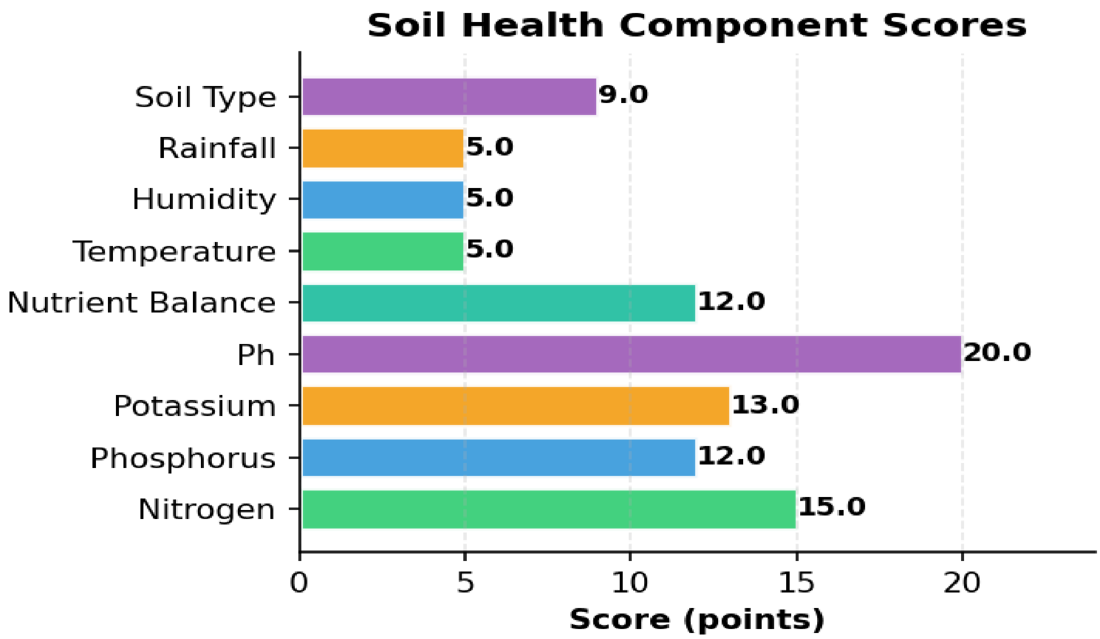
verall Soil Health Score	96.0/100	Excellent

Component	Score	Points
Nitrogen	15.0	points
Phosphorus	12.0	points
Potassium	13.0	points
Ph	20.0	points
Nutrient Balance	12.0	points
Temperature	5.0	points
Humidity	5.0	points
Rainfall	5.0	points

Soil Health & Crop Recommendation System

■ Powered by ML + IoT + Weather Intelligence

Soil Type	9.0	points



Weather Suitability for Pigeonpeas

Overall Suitability Score: 7.3/10 ■ Good

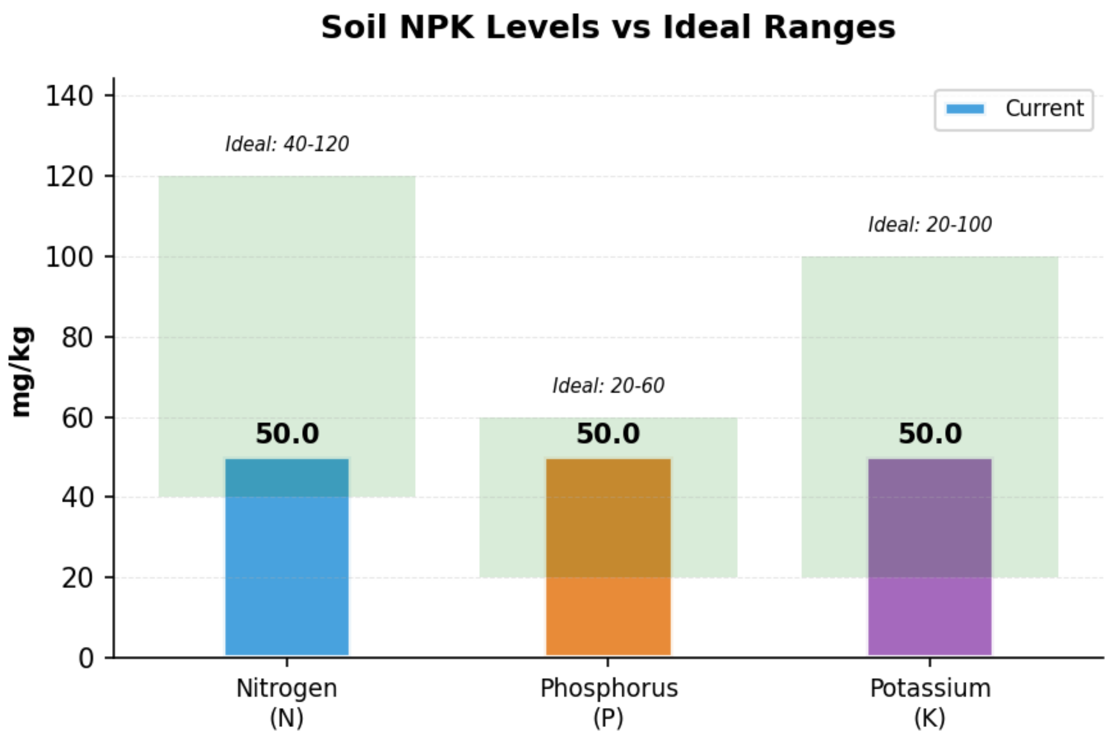
Weather conditions are good for this crop

- ✓ Temperature 25.0°C is ideal for pigeonpeas
- ✓ Humidity 60.0% is ideal for pigeonpeas — reduces disease risk
- ■ Rainfall 120 mm is below requirement (600-1000 mm); irrigation strongly recommended

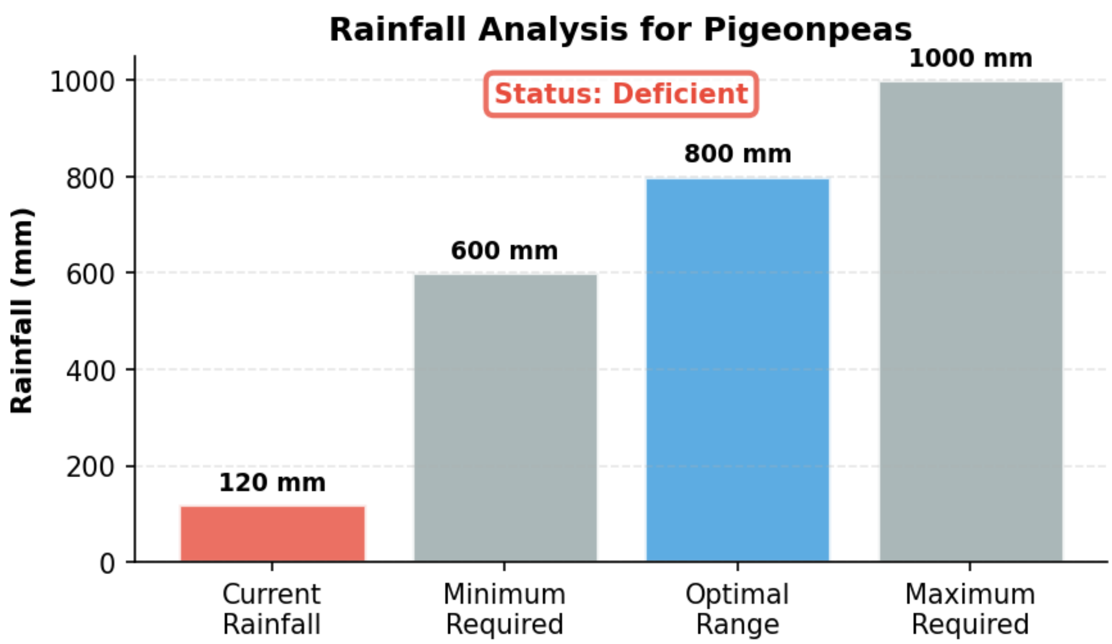
Recommended Crop

Predicted Crop	Pigeonpeas
Confidence Level	82.9%
Model Used	Stacking Classifier

Soil NPK Levels Visualization



Rainfall Analysis



■■ Warnings & Observations

Action Required:

- ■ Rainfall below optimal → Increase irrigation

Positive Indicators:

- ✓ Nitrogen level optimal
- ✓ Phosphorus level optimal
- ✓ Potassium level optimal
- ✓ pH optimal (6.5)
- ✓ Soil type: Black Soil

■ Market Price & Profitability Analysis

Price Range	2000 - 3500 ■/Quintal
Average Price	2750 ■/Quintal

Best Market: Local Market, Your State (2750 ■/Quintal)

Market Advice: Market data integration is available in the system. In a production environment, this would show real-time market prices.

■ Soil Treatment Recommendations

- Soil pH (6.5) is within optimal range (6.0-7.5) for pigeonpeas.
- Nitrogen (50.0 mg/kg) is excessive for pigeonpeas. Optimal range: 10-30. Reduce nitrogen application to avoid excessive vegetative growth.
- Phosphorus (50.0 mg/kg) is within optimal range for pigeonpeas.
- Potassium (50.0 mg/kg) is excessive for pigeonpeas. Optimal range: 10-30. High K can cause magnesium deficiency.
- High clay content, retains moisture well. Ideal for cotton, wheat, and pulses.

■ Fertilizer Recommendations

FARM DETAILS

Land Size: 1.0 acres

Current Soil NPK: N=50.0 | P=50.0 | K=50.0 mg/kg

Optimal NPK for pigeonpeas: N=20 | P=50 | K=20 kg/acre

SOIL NUTRIENT STATUS

✓ Excellent: Soil NPK levels are already optimal for pigeonpeas

Maintain soil health through:

- Regular organic compost application
- Crop rotation practices
- Green manuring between seasons

ORGANIC & BIO-FERTILIZER ALTERNATIVES

Farm Yard Manure (FYM): 5-10 tons

■■ Application: 5-10 tons/acre (provides balanced NPK)

Compost: Well-decomposed organic compost

■■ Application: 3-5 tons total

Green Manure: Dhaincha, Sunhemp, or other leguminous crops

■■ Benefits: Natural nitrogen fixation, improves soil structure

Bio-Fertilizers:

- Rhizobium: 200g/acre (for nitrogen fixation)
- PSB (Phosphate Solubilizing Bacteria): 200g/acre
- Azospirillum: 200g/acre (for nitrogen fixation)
- VAM (Vesicular Arbuscular Mycorrhiza): Enhances P uptake

Note: Apply fertilizers based on soil test results and local
agricultural extension recommendations for best results.

■ Crop-Specific Tips for Pigeonpeas

Drought-tolerant legume. Requires well-drained soil and warm climate.

■ Additional Notes & Disclaimer

- These recommendations are based on comprehensive soil analysis and advanced ML predictions.
- Actual results may vary based on local conditions, weather patterns, and farming practices.
- Consult with local agricultural experts before making major farming decisions.
- Regular soil testing (every 2-3 years) is recommended for optimal crop management.
- Consider crop rotation and organic farming practices for long-term soil health.
- Monitor weather forecasts and adjust irrigation schedules accordingly.